

OCR Further Mathematics
Mechanics - Work, Energy and Power
Mark Scheme
AS and A Level
Further Mathematics A - H235, H245



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Question			Answer/Indicative content	Marks	Part marks and guidance		
1	a		$20 = 4u \Rightarrow u = 5$ Initial energy = $\frac{1}{2} \times 4 \times 5^2$ Energy at $\theta = \frac{1}{2} \times 4 \times v^2 + 4g \times 0.8(1 - \cos\theta)$ $2v^2 + 15.68 = 50 \Rightarrow v^2 = 17.16$ Radial: $a_r = \frac{v^2}{0.8} = \frac{17.16}{0.8}$ Tangential: $ma_t = -mg \sin \frac{\pi}{3}$ $a = \sqrt{\left(-\frac{\sqrt{3}g}{2}\right)^2 + \left(\frac{429}{20}\right)^2} = 23.067...$ so the magnitude of the acceleration is 23.1 m s^{-2} (3 sf)	B1 (AO1.1) B1 (AO1.1) M1 (AO1.1) A1 (AO1.1) M1 (AO3.1b) M1 (AO3.1b) A1 (AO1.1) [7]	= 50 Attempt to derive total ME at general or specific angle Equating energies to derive a value for v^2 Correct form for centripetal acceleration and use of v^2 NII for tangential direction with weight resolved (– not necessary)	Assuming zero PE level at initial level of P $v = 4.142...$ $a_r = 21.45$ $a_t = -\frac{\sqrt{3}g}{2} = -8.4870...$	
	b		Radial: $T - 4g \cos\theta = \frac{4v^2}{0.8}$ $v^2 = 5^2 - 2g \times 0.8(1 - \cos\theta)$ $-7.84 \cos\theta = 9.32 + 15.68 \cos\theta$ $\therefore \cos\theta = -\frac{9.32}{23.52}$ $\therefore \theta = 113.3^\circ \text{ or } 1.98 \text{ rads}$	M1 (AO2.1) M1 (AO2.1) A1 (AO3.2a) [3]	NII for radial direction. T could be set to 0. Correct form of a_r . v^2 in terms of from conservation of energy	$v^2 = 9.32 + 15.68 \cos\theta$	
			Total	10			
2			Initial Elastic PE = $\frac{24 \times 0.9^2}{2 \times 0.6}$	B1 (AO1.1)	Use of	16.2 J	



Question			Answer/Indicative content	Marks	Part marks and guidance		
			Final Elastic PE = $= \frac{24 \times 0.4^2}{2 \times 0.6}$	B1 (A01.1)	$\frac{\lambda x^2}{2l}$ with	3.2 J	
			Increase in PE = $0.4g \times 2.5$	M1 (A01.1)	attempt at finding extension (ie not just $x = 1.5$)	9.8 J	
			"16.2" = "3.2" + $\frac{1}{2} \times 0.4v^2$ + "9.8"	M1 (A01.1)			
			$v^2 = 16 \Rightarrow$ speed is 4 ms^{-1}	A1 (A01.1) [5]	Use of $\frac{\lambda x^2}{2l}$ with attempt at finding extension (ie not just $x = 1$) Attempt at use of "mgh" to find the increase of gravitational PE from initial position to ceiling Attempt at conservation of energy with consideration of KE and their PE Not ±. Units required.	8.624 J	
			Total	5			



Question			Answer/Indicative content	Marks	Part marks and guidance		
3	a		$T = \frac{\lambda x}{l}$ and $r = l + x$ both used in solution $T = \frac{mv^2}{l + x}$ $\frac{\lambda x}{l} = \frac{mv^2}{l + x} \Rightarrow \lambda x(l + x) = lmv^2$ $\Rightarrow \lambda x^2 + \lambda lx - lmv^2 = 0$	M1 (AO3.3) M1 (AO3.3) A1 (AO3.3) [3]	Use of $F = ma$ with centripetal acceleration AG		
	b		$x = \frac{-\lambda l + \sqrt{(\lambda l)^2 - 4\lambda(-lmv^2)}}{2\lambda}$ $x = \frac{l}{2} \left(\left(1 + \frac{4mv^2}{\lambda l} \right)^{\frac{1}{2}} - 1 \right)$ $x = \frac{l}{2} \left(1 + \frac{1}{2} \frac{4mv^2}{\lambda l} + \dots - 1 \right)$ $x \approx \frac{l}{2} \left(\frac{1}{2} \frac{4mv^2}{\lambda l} \right) = \frac{mv^2}{\lambda} \Rightarrow \lambda x \approx mv^2$	M1 (AO2.1) M1 (AO3.1b) A1 (AO1.1) [3]	Use of the quadratic equation formula Reject negative route and rearranging to form with $\sqrt{1 + \dots}$ Use of binomial series AG	No need to mention $\left \frac{4mv^2}{\lambda l} \right < 1$	
	c		$\lambda x \approx mv^2 \Rightarrow \frac{\lambda x}{l} \approx T \approx \frac{mv^2}{l}$ and if the string were inextensible, corresponding to an infinite value of λ and x being 0, then l would be the radius of the motion and so the RHS would be the centripetal force	B1 (AO3.5a) [1]			
	d		$v \approx \sqrt{\frac{260 \times 0.03}{0.16}} = \text{awrt } 7.0$	B1 (AO3.4) [1]	(6.982...)		



Question			Answer/Indicative content	Marks	Part marks and guidance	
	e		While v in this situation is slightly below 7 nevertheless it is an estimate so we cannot be certain that the modelled value does not exceed 7 in which case the assumption would not be justified	B1 (AO2.2b) [1]		
			Total	9		



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Question		Answer/Indicative content	Marks	Part marks and guidance		
4	a	$\frac{1}{2} \times 2.8 \times 6^2 + 2.8 \times 9.8(0.75 - 0.75 \cos \theta) = \frac{1}{2} \times 2.8 v^2$ $50.4 + 20.58 - 20.58 \cos \theta = 1.4 v^2$ $v^2 = 50.7 - 14.7 \cos \theta$	M1 (AO3.3) 1.1) M1 A1 (AO2.1) [3]	Attempt at GPE 0.75cos2.8g with correct theta and length Conservation of energy, allow sign errors AG An intermediate step must be shown	In this solution θ is the angle between OA and the upwards vertical through O	
	b	$T + 2.8 \times 9.8 \cos \theta = 2.8 \times \frac{v^2}{0.75}$ $200 + 2.8 \times 9.8 \cos \theta = 2.8 \times \frac{50.7 - 14.7 \cos \theta}{0.75}$ $\theta = 97.5^\circ \text{ or } 1.70 \text{ rad}$	M1 (AO3.3) A1 (AO1.1) M1 (AO2.2a) A1 (AO1.1) [4]	Using $F = ma$ in radial direction with tension, resolved weight and an expression for centripetal acceleration Using their expression for v^2 and breaking tension of string $\cos \theta = -\frac{134}{1029} = -0.1802...$	Allow sin instead of cos	
		Total	7			



Question		Answer/Indicative content	Marks	Part marks and guidance		
5		Initial energy = $80g(x + 4)$ where x is the final extension of the rope Elastic PE = $\frac{470x^2}{2 \times 4}$ tic Energy lost (work done by air resistance) = $32(x + 4)$ $80g(x + 4) = \frac{470x^2}{2 \times 4} + 32(x + 4)$ $235x^2 - 3008x - 12032 = 0$ soi $x = 16$ She falls 20 m before first coming to rest	M1 (A03.3) M1 (A03.3) M1 (A03.3) A1 (A03.4) M1 (A01.1) A1 (A02.3) [6]	Attempt GPE Attempt EPE Attempt Work done Initial energy = Final Elastic PE + energy lost Can be multiple of this Only from W-E principle with three terms -3.2 rejected	Or $80gd$ where d is the distance below the bridge $\frac{470(d - 4)^2}{2 \times 4}$ $80gd = \frac{470(d - 4)^2}{8} + 32d$ $235d^2 - 4888d + 3760 = 0$ $d = 20$ (and 0.8 rejected)	
Total			6			

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Question			Answer/Indicative content	Marks	Part marks and guidance		
6	a		$\begin{pmatrix} 2 \\ 10 \end{pmatrix} \cdot \begin{pmatrix} 50 \\ 140 \end{pmatrix}$ $2 \times 50 + 10 \times 140 = 1500 \text{ J}$	M1 (AO1.1) A1 (AO1.1) [2]	Using WD = F.x		
	b		1500/5 = 300 W	B1ft (AO1.1a) [1]	Their 1500	Must be a scalar value	
	c		$\frac{1}{2} \times 1.25 v^2 = \frac{1}{2} \times 1.25 \times 10^2 + 1500$ 50 ms^{-1} Alternative method $\begin{pmatrix} 50 \\ 140 \end{pmatrix} = 5\mathbf{u} + \frac{1}{2} \cdot \begin{pmatrix} 8 \\ 5 \end{pmatrix} \cdot 5^2 \Rightarrow \mathbf{u} = \begin{pmatrix} 6 \\ 8 \end{pmatrix}$ The $\mathbf{v} = \begin{pmatrix} 6 \\ 8 \end{pmatrix} + \begin{pmatrix} 8 \\ 5 \end{pmatrix} \cdot 5 = \begin{pmatrix} 14 \\ 48 \end{pmatrix}$ n $ \mathbf{v} = 50 \text{ ms}^{-1}$	M1 (AO1.1) A1 (AO1.1) M1 A1 [2]	Correct use of work-energy principle Not ± complete method involving constant acceleration formula(e)	Can use their WD for M mark	
			Total	5			



Question		Answer/Indicative content	Marks	Part marks and guidance		
7	a	$\frac{1}{2}mv^2 + m\gamma r \cos \theta = m\gamma r \cos \alpha_{oe}$ $v^2 + 2\gamma r \cos \theta = \frac{3}{2}\gamma r$ $m\gamma \cos \theta (-C) = ma$ $a = \frac{v^2}{r}$ $\cos \theta = \frac{a}{\gamma} = \frac{v^2}{\gamma r}$ $v^2 = \frac{1}{2}\gamma r$ $r \cos \theta = \left(\sqrt{\frac{1}{2}\gamma r \sin \theta} \right) t + \frac{1}{2}\gamma t^2$ $t^2 + \sqrt{\frac{3r}{2\gamma}}t - \frac{r}{\gamma} = 0$ $t = \frac{1}{4}\sqrt{\frac{r}{\gamma}}(\sqrt{22} - \sqrt{6})$ $OF = r \sin \theta + \sqrt{\frac{1}{2}\gamma r \cos \theta} \times \frac{1}{4}\sqrt{\frac{r}{\gamma}}(\sqrt{22} - \sqrt{6})$ $OF = \frac{r}{8}(\sqrt{11} + 3\sqrt{3})_{oe}$	M1(A03.3) A1(A01.1) B1(A01.1) B1(A01.1) M1(A02.2a) A1(A02.2a) M1(A03.4) *M1(A01.1) Dep *M1(A01.1) Dep *M1(A03.4) A1(A01.1) [11]	Conservation of energy Nil for P at point where it is about to lose contact with surface. soi Use previous two results to relate v and $\cos \theta$ for v or v^2 or $\cos \theta = \frac{1}{2}$ Use of $s = ut + \frac{1}{2}at^2$ using their vertical component of v as u where v has come from consideration of theta Reduction to 3 term quadratic with numerical values for trig ratios Solve for t (BC)	θ is the angle between OP and the upward vertical Could see contact force, C, later set to 0 Or use of trajectory eqn: $y = x \tan \theta + \frac{\gamma x^2}{2v^2 \cos^2 \theta}$ with $y = r$ $\cos \theta$ and $v^2 = \frac{1}{2}\gamma r$ $8x^2 + 2\sqrt{3}rx - r^2 = 0$ $x = \frac{\sqrt{11} - \sqrt{3}}{8}r$ $r \sin \theta + \frac{\sqrt{11} - \sqrt{3}}{8}r$	



Question			Answer/Indicative content	Marks	Part marks and guidance	
					Find OF	
					<u>Examiner's Comments</u> Candidates found this question difficult but most candidates were given some marks. Higher ability candidates drew good diagrams and considered conservation of energy between the start point and the point where the particle leaves the surface. Newton's Second Law was then used to find $\cos \theta$ and an expression for v . These candidates were given 6 marks for doing this successfully. Not many candidates were given any further marks for considering vertical motion to find an expression for t and then considering horizontal motion (including adding on $r \sin \theta$). In this second part of the question candidates who took time to organise their projectile equations generally made better progress. This was a challenging question where the best candidates were able to produce a detailed and well-structured solution. Some candidates set the start point at the top of the hemisphere and that the particle left the hemisphere at α which made it very difficult for them to be given any marks. Incorrect signs were used in the conservation of energy equation (mainly from	

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Question			Answer/Indicative content	Marks	Part marks and guidance	
					candidates having difficulties understanding their zero level of potential energy) and other candidates didn't introduce another angle (i.e. θ) where the particle left the surface.	
	b		Unchanged since OF does not depend on γ	E1(A03.5a) [1]	<u>Examiner's Comments</u> If candidates had worked out part (a) correctly they were able to argue that OF didn't depend on gravity so it would remain unchanged.	
			Total	12		



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Question		Answer/Indicative content	Marks	Part marks and guidance		
8	a	$\cos \theta = \frac{1}{2}$ or $\sin \theta = \frac{\sqrt{3}}{2}$ $v \sin \theta = (1+m)V$ $V = \frac{\sqrt{3}v}{2(1+m)}$ $I = mV$ or $I = \pm(V - v \sin \theta)$ $V = \frac{v \times \frac{1}{2} \sqrt{3}}{1+m} \Rightarrow I = \frac{\sqrt{3}mv}{2(1+m)}$	B1(A03.1b) M1(A03.1b) A1(A03.1b) A1(A02.2a) [4]	Must be clear where it comes from ie. Shown in diagram or clear use of distances. May be other way round Conservation of momentum in direction of string Working must be seen AG Justification of using Impulse = change in momentum	θ is the angle between the string and the direction of AB when the string initially becomes taut	



Question			Answer/Indicative content	Marks	Part marks and guidance	
					to find an equation for the final velocity in terms of m and v. Most candidates were unable to show logical reasoning and seemed to use the answer given in the question to form their answer. For the final mark examiners had to be certain that the candidate had considered change of momentum on either particle. Just multiplying their expression for velocity by 'm' wasn't enough to demonstrate this. A significant amount of candidates were unable to access this question because they were trying to model the problem in 1 dimension.  AFL When answering a question (particularly 'show that') encourage candidates to write what they are using to set up the equation and in what direction to communicate clearly what the intention is. For instance: 'CoM parallel to the string' or 'Impulse on A'	



Question		Answer/Indicative content	Marks	Part marks and guidance		
	b	$KE = \frac{1}{2}(v^2 + (v \cos \theta)^2)_{\text{soi}}$ $KE = \frac{1}{2} \left(\left(\frac{\sqrt{3}v}{2(1+m)} \right)^2 + \left(v \times \frac{1}{2} \right)^2 \right)$ $KE = \frac{v^2(4 + 2m + m^2)}{8(1+m)^2}$	M1(A03.1b) M1(A01.1) A1(A01.1) [3]	Transverse component unchanged and using their longitudinal component Substituting in for V and $\cos \theta$ (could just be in speed equation) Or any equivalent single algebraic fraction	Condone consideration of speed squared or inclusion of m in KE	<u>Examiner's Comments</u> This was the most difficult question on the paper for candidates to access. To be given the first 2 marks candidates needed to identify the 2 components of B's final speed and square each of them. Out of the minority of candidates who managed this, the main error was to forget that $m = 1$ in the kinetic energy formula leading to an extra m in their final answer.



Question			Answer/Indicative content	Marks	Part marks and guidance		
	c		$V = \frac{\sqrt{3}v}{2(1+m)} \rightarrow 0 \text{ as } m \rightarrow \infty$ <i>B</i> moves (approximately) in a circle around <i>A</i>	B1(A03.2a) B1(A02.2b) [2]	If <i>m</i> is very large then <i>A</i> is approximately stationary or <i>B</i> has only its transverse velocity of $\frac{1}{2}v$ after the string becomes taut <u>Examiner's Comments</u> Some candidates omitted this question after struggling with the first 2 parts; however this question was accessible if candidates had modelled the situation. Essentially if <i>m</i> is very large, then <i>A</i> will be stationary as <i>B</i> will be unable to move it. As <i>A</i> is a fixed point then <i>B</i> will travel in circular motion around it. Some candidates were able to infer circular motion but didn't refer to <i>A</i> being stationary. The most common incorrect answers were that <i>B</i> would stop or performing simple harmonic motion.		
			Total	9			

$T\cos\theta + R\sin\theta = mg$	*M1(A03.3)	m, g, r and / or T may appear as	Must have 2
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Question			Answer/Indicative content	Marks	Part marks and guidance		
			$v = 2.48$ Alternate method $\theta = 36.9$ (or 53.1) $r(= 3.5 \sin\theta) = 2.1$ Acc component parallel to string is $a \sin \theta$ $T - mg \cos \theta = ma \sin \theta$ $= m \frac{v^2}{r} \sin \theta$ $v^2 = \frac{r(T - mg \cos \theta)}{m \sin \theta}$ $v = 2.48$	A1(AO1.1) B1 B1 M1 M1 M1 M1 A1 [7]	and contact force $Us^a = \frac{v^2}{r}$ to e eliminate a and introduce v Solve simultaneously to eliminate R Substitute values and find v Or correct trig ratio given Candidate may give θ between horizontal so cos and sin values will be the other way round soi m, g, r and / or T may appear as values throughout Attempt to resolve acceleration NII with 3 terms, parallel to string	well May find $R = \frac{53}{6}$ and sub in Or $\frac{\sqrt{154}}{5}$ θ is the angle between the string and the downwards vertical Must have 3 terms; allow sign / trig errors	



Question			Answer/Indicative content	Marks	Part marks and guidance	
					Us $a = \frac{v^2}{r}$ to e eliminate a and introduce v Rearrange to make v^2 the subject or solve for v <u>Examiner's Comments</u> The majority of candidates were able to get some marks on this question, although few received full marks. Pretty much all the candidates were able to find the trig ratios and use $a = \frac{v^2}{r}$ for 2 marks. Fewer candidates used the correct value for the radius. Some candidates were able to set up equations to solve. Common mistakes were a lack of understanding of resolving components and a misunderstanding of the direction of acceleration. Some candidates discounted R or did not resolve it. Candidates who didn't draw a good force diagram significantly disadvantaged themselves here.	
			Total	9		



Question		Answer/Indicative content	Marks	Part marks and guidance		
10	a	$\mathbf{x}(5) = 5 \begin{pmatrix} 2 \\ -5 \end{pmatrix} + \frac{1}{2} \times 5^2 \times \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ $\mathbf{x}(5) = \begin{pmatrix} 22.5 \\ 0 \end{pmatrix}$ $\mathbf{v.v} = \begin{pmatrix} 2 \\ -5 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -5 \end{pmatrix} + 2 \begin{pmatrix} 1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 22.5 \\ 0 \end{pmatrix}$ $\mathbf{v.v} = 4 + 25 + 45 = 74$ $KE = \frac{1}{2} \times m \times 74 = 37m$ Alternate method Substitute $\mathbf{x} = \mathbf{ut} + \frac{1}{2} \mathbf{at}^2$ into $\mathbf{v.v} = \mathbf{u.u} + 2\mathbf{a.x}$ to get $\mathbf{v.v} = \mathbf{u.u} + 2\mathbf{a.ut} + \mathbf{aat}^2$ $\mathbf{v.v} = \begin{pmatrix} 2 \\ -5 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -5 \end{pmatrix} + 2 \times 5 \begin{pmatrix} 1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -5 \end{pmatrix} + 25 \begin{pmatrix} 1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ $\mathbf{v.v} = 4 + 25 + 10(2 - 10) + 25(1 + 4) = 74$ $KE = \frac{1}{2} \times m \times 74 = 37m$	M1(A03.1b) A1(A01.1) M1(A03.1b) M1(A01.1) A1(A02.2a) M1 A1 M1 M1 A1 [5]	soi AG www or v.v with $\mathbf{v} = \mathbf{u} + \mathbf{at}$ AG	Award SC2 for using $\mathbf{v} = \mathbf{u} + \mathbf{at}$ to find $\mathbf{v}$ and then KE Must be complete and correct if not SC0	

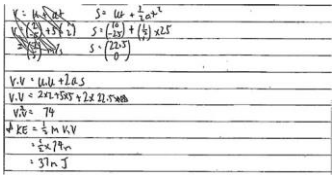
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Examiner's Comments

Most candidates achieved 3 or more marks on this question. Some students didn't 'use' the equation given to find the kinetic energy which resulted in them only being given 2 marks – candidates need to be aware that they must use the equation given in this type of question. As detailed in the specification, in a 'show that' question, the explanation has to be sufficiently detailed to get



Question			Answer/Indicative content	Marks	Part marks and guidance	
					full marks. Examiners wanted to see values substituted into the equation and the dot product applied correctly (the dot product gives a scalar value and not a vector). Exemplar 1 	
	b	i	$P = m \begin{pmatrix} 1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -5 \end{pmatrix} \text{ or equiv vector form}$ $p = m(2-10) = -8m$	M1(A02.2a) A1(A01.1) [2]	AG www Must see 2-10 or 2m -10m <u>Examiner's Comments</u> Again, most candidates were able to answer this question but some lost a mark for not giving a detailed enough method. For full marks it was important to see the values substituted in, the dot product calculated and then the final answer.	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	<i>E.g. Q is slowing down</i>	B1(A02.2a) [1]	or Kinetic Energy is decreasing <u>Examiner's Comments</u> Most candidates answered this incorrectly by stating the misconception below. Some candidates were able to understand that the force is opposing the motion, however neglected to translate this to a loss of speed or kinetic energy. ? Misconception Candidates thought that negative power means that the particle is 'travelling backwards'.	



Question			Answer/Indicative content	Marks	Part marks and guidance		
	c	i	$\mathbf{v} = \begin{pmatrix} 2 \\ -5 \end{pmatrix} + \begin{pmatrix} 1 \\ 2 \end{pmatrix} t \text{ or}$ $\mathbf{v} = \begin{pmatrix} 2+t \\ -5+2t \end{pmatrix}$ $m\mathbf{a} \cdot \mathbf{v} = m \begin{pmatrix} 1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 2+t \\ -5+2t \end{pmatrix} = 0 \Rightarrow 2+t+2(-5+2t) = 0$ Time taken is 1.6 s Alternate method $\mathbf{v} = \begin{pmatrix} 2 \\ -5 \end{pmatrix} + \begin{pmatrix} 1 \\ 2 \end{pmatrix} t \text{ or}$ $\mathbf{v} = \begin{pmatrix} 2+t \\ -5+2t \end{pmatrix}$ $\frac{1}{2} m \mathbf{v} \cdot \mathbf{v} = \frac{1}{2} m (5t^2 - 16t + 29) \Rightarrow 10t - 16 = 0$ Time taken is 1.6 s	B1(AO1.1) M1(AO2.2a) A1(AO1.1) B1 M1 A1 [3]	$\mathbf{v} = \mathbf{u} + \mathbf{a}t$ soi Allow missing m; must have $= 0$ www $\mathbf{v} = \mathbf{u} + \mathbf{a}t$ soi Find minimum www	May be derived from integrating a Must be equation in terms of t. Condone one slip Allow missing $\frac{1}{2}m$	

Examiner's Comments

The majority of candidates answered this well. The most common error was to set the velocity to 0. Some candidates forced the velocity vector to be perpendicular to the acceleration vector which was a lot longer method. Candidates were more prone to making errors using this method.

As the direction was 'Find' candidates didn't have to be as thorough with their methods as in the first two parts of this question, but accurate working allows for partial credit if mistakes made, and helps avoid the risk of making mistakes.



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\frac{1}{2}mv.v = \frac{1}{2}m \left(\frac{2+1.6}{-5+2 \times 1.6} \right) \cdot \left(\frac{2+1.6}{-5+2 \times 1.6} \right)$ $\frac{81}{10} \text{ m J or } 8.1 \text{ m J}$	M1(A02.2a) A1(A01.1) [2]	With their 1.6 <u>Examiner's Comments</u> Most of the candidates who answered Q3 (c)(i) correctly were able to access this question. Some candidates didn't realise that they could use the time at which the power is zero to find the minimum kinetic energy, and instead spent time restarting the solution.	
			Total	13		



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Question			Answer/Indicative content	Marks	Part marks and guidance		
11		a	Use $\bar{x} = \frac{\sum mx}{\sum m}$ or $\bar{y} = \frac{\sum my}{\sum m}$ of $\bar{x} = \frac{2 \times 0 + 3 \times 0.6 + 5 \times 0.4}{2 + 3 + 5} = \frac{3.8}{10} = 0.38$ (AG) $\bar{y} = \frac{2 \times 0 + 3 \times 0 + 5 \times 0.2}{2 + 3 + 5} = \frac{1}{10} = 0.1$	M1 (AO 1.2) A1 (AO 3.3) A1 (AO 3.3) [3]	Values must be substituted Intermediate step must be seen		
		b	G is not on the frame so the weight would have an unbalanced moment about the support point	E1 (AO 2.4) [1]			
		c	Moments about B: $0.6T = 10g \times 0.1$ $T = \frac{g(0.8-l)}{l}$ Eliminate $T: \frac{5g}{3} = \frac{g(0.8-l)}{l} \Rightarrow 8l = 2.4 \Rightarrow l = 0.3$ (AG)	M1 (AO 3.4) M1 (AO 3.4) A1 (AO 1.1) [2]	Each term must be of the form $F \times d$ Correct form for Hooke's law Intermediate working must be seen		



Question			Answer/Indicative content	Marks	Part marks and guidance	
		d	Any two resolving equations oe, eg: (vertically): $R \cos \theta + F \sin \theta = 10g$ (horizontally): $R \sin \theta = F \cos \theta + T$ (parallel to plane): $F \cos \theta + T = 10g \sin \theta$ (perpendicular to plane): $R = T \sin \theta + 10g \cos \theta$ Use $T = \frac{5}{3}g$, $\sin \theta = \frac{3}{5}$, $\sin \theta = \frac{3}{5}$ to derive equations $4R + 3F = 50g$ and $9R = 12F + 25g$ oe $R = 9g$ and $F = \frac{14}{3}g$ $F \leq \mu R \Rightarrow \frac{14}{3}g \leq \mu \times 9g$ $\therefore \mu \geq \frac{14g}{3 \times 9g} = \frac{14}{27}$ (AG)	M1 (AO 3.4) M1 (AO 3.4) M1 (AO 1.1a) A1 (AO 1.1) M1 (AO 3.4) A1 (AO 1.1) [6]	Or taking moments eg about the point where the string is attached to the plane ($1 \times N = (0.8 + 0.1) \times 10g$) or the point where the lines of action of R and the weight intersect: $\frac{0.1}{\sin \theta} \times F = \left(0.6 - \frac{0.1}{\tan \theta}\right) \times T$ Forming equations that lead to numerical values for F and R Both $F \leq \mu R$ with values substituted in Intermediate working must be seen	R is the normal contact force, F is the frictional force (up the slope)
			Total	13		



Question			Answer/Indicative content	Marks	Part marks and guidance	
12		a	$15\,000 \times 6 = \frac{1}{2} \times 800 v^2$ $v = \sqrt{225} = 15 \text{ m s}^{-1}$	M1 (AO 1.1a) A1 (AO 1.1) [2]	Finding energy input using $W = Pt$ and equating to KE gained $\frac{15\,000}{v} = 800 \frac{dv}{dt}$ Or $\Rightarrow \frac{75}{2} \int_0^6 dt = \int_0^v 2v \, dv$ oe	
		b	Considering forces along the road: $800 g \sin \theta + 150 = D$ $15\,000 = Dv$ $v = \frac{15\,000}{800g \times \frac{1}{20} + 150} = \text{awrt } 27.7 \text{ ms}^{-1}$	M1 (AO 1.1a) M1 (AO 1.1) A1 (AO 1.1) [3]	Where D is the driving force Use of $P = Fv$ 27.675276	Or $W = 800 g d \sin \theta + 150d$, where d is distance travelled in time t and $W = 15\,000t$ and $v = \frac{d}{t}$
		c	$D = \frac{15000}{30} \quad (=500)$ $\frac{15\,000}{30} = R_{\text{total}}$ $R_c = \frac{3}{4} \times 500 = 375 \text{ N}$ Car: $500 = T + 375 \Rightarrow T = 125 \text{ N}$	M1 (AO 1.1a) M1 (AO 2.2a) A1 (AO 1.1) A1 (AO 1.1) [4]	Driving force equals total resistance Or Trailer: $T = R_T = \frac{1}{4} \times 500 = 125 \text{ N}$	
			Total	9		

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Question			Answer/Indicative content	Marks	Part marks and guidance		
14		a	$\frac{1}{2}mu^2 = mgh(1 - \cos \theta)$ $mu = 2mv_B$ $e = \frac{v_B}{u}$ $v_B = \frac{1}{2}u$ and $e = \frac{1}{2}$ $2m \times \frac{1}{2}u = mV_A + 2mV_B$ $\frac{1}{2} = \frac{V_A - V_B}{\frac{1}{2}u}$ $V_A = \frac{1}{2}u$ $\frac{1}{2}m\left(\frac{1}{2}u\right)^2 = mgh(1 - \cos \phi)$ $\frac{1}{8}mu^2 = \frac{1}{4}mgh(1 - \cos \theta) = mgh(1 - \cos \phi)$ $\Rightarrow \cos \phi = \frac{3 + \cos \theta}{4}$	M1 (AO 3.3) M1 (AO 3.4) M1 (AO 1.1) A1 (AO 1.1) M1 (AO 3.4) M1 (AO 3.4) A1 (AO 1.1) M1 (AO 3.4) A1 (AO 2.1) [9]	Conservation of energy for A Conservation of momentum Restitution Conservation of momentum for 2nd collision with unchanged v_B Restitution for 2nd collision Conservation of energy for A AG so elimination of $\frac{1}{2}mu^2$ or must be clearly shown	v_B is velocity of B after first collision	

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Question			Answer/Indicative content	Marks	Part marks and guidance		
		b	$V_B = \frac{1}{4}u$ $2m \times \frac{1}{4}u + m \times \left(-\frac{1}{2}u\right) = mv_A + 2mv_B$ $\frac{1}{2} = \frac{v_A - v_B}{\frac{1}{4}u - \left(-\frac{1}{2}u\right)}$ $v_B = -\frac{1}{8}u \text{ and } v_A = \frac{1}{4}u$ $\text{KE loss} = \frac{\frac{1}{2}m \times \left(\frac{1}{4}u\right)^2 + \frac{1}{2} \times 2m \times \left(-\frac{1}{8}u\right)^2}{\frac{1}{2}mu^2} = \frac{3}{32} = 9.4\%$ So percentage of KE remaining is 90.6%	B1 (AO 1.1) M1 (AO 3.4) M1 (AO 3.4) A1 (AO 1.1) A1 (AO 1.1)	Conservation of momentum for 3rd collision Restitution for 3rd collision	Or with signs reversed, if different 'positive' direction chosen	
		c	During and between the 1st and 2nd collisions there is no external horizontal force acting on A and while there is an external force acting on B its velocity and hence momentum is the same just before the 2nd collision as it is just after the 1st collision	E1 (AO 2.4) [1]	Identifying the unchanged velocity for B in the two collisions		
		d	Between the 2nd and 3rd collisions a component of the tension in the string OA is horizontal and so the momentum of the system is not conserved	E1 (AO 2.4) [1]	Identifying the relevant external force acting		
			Total	16			



Question			Answer/Indicative content	Marks	Part marks and guidance		
15		a	$T - kv^2 = mv \frac{dv}{dx}$ $m \int \frac{-2kv}{T - kv^2} dv = \int -2k dx$ $m \ln(T - kv^2) = -2kx + c$ $\ln(T - kv^2) = \frac{-2kx + c}{m} \Rightarrow T - kv^2 = e^{\frac{-2kx + c}{m}}$ $\Rightarrow T - kv^2 = e^{\frac{-2kx}{m}} e^{\frac{c}{m}} \Rightarrow v^2 = \frac{1}{k} \left(T - A e^{\frac{-2kx}{m}} \right)$	M1 (AO 3.3) M1 (AO 3.4) M1 (AO 1.1) M1 (AO 1.1) A1 (AO 2.1) [5]	NII with 3 terms and $a = v \frac{dv}{dx}$ used Separating the variables Correctly integrating both sides Exponentiation, giving $f(v) = e^{g(x)}$ AG so intermediate step needed	Condone sign errors and/or missing m oe, eg stating $A = e^{\frac{c}{m}}$	



Question			Answer/Indicative content	Marks	Part marks and guidance		
		b	Work done by constant force is $T(1 - 0) = T$ $x = 0, v = 0$ gives $v^2 = \frac{T}{k} \left(1 - e^{-\frac{2kx}{m}} \right)$ KE gain is $\frac{1}{2}mv_1^2 = \frac{mT}{2k} \left(1 - e^{-\frac{2k}{m}} \right)$ So work done against resistance $= T - \frac{mT}{2k} \left(1 - e^{-\frac{2k}{m}} \right)$ Alternative solution $x = 0, v = 0$ gives $v^2 = \frac{T}{k} \left(1 - e^{-\frac{2kx}{m}} \right)$ WD against resistance $= \int_0^1 kv^2 dx = T \int_0^1 \left(1 - e^{-\frac{2kx}{m}} \right) dx$ $= T \left[x + \frac{m}{2k} e^{-\frac{2kx}{m}} \right]_0^1$ $= T \left(1 + \frac{m}{2k} e^{-\frac{2k}{m}} - \frac{m}{2k} \right)$	B1 (AO 1.1) M1 (AO 3.4) M1 (AO 3.4) A1 (AO 3.4) M1 M1 M1 A1 [4]	Use of initial conditions to find A oe Use of initial conditions to find A oe		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		c	The limiting velocity is $\sqrt{\frac{T}{k}}$	B1 (AO 3.2a) [1]		
			Total	10		



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Question			Answer/Indicative content	Marks	Part marks and guidance		
16		a	(i) $T_A = \frac{\lambda_A \times 2}{2}$ and $T_B = \frac{\lambda_B \times 1}{2}$ $\cos PAB = \frac{4}{5}$ and $\cos PBA = \frac{3}{5}$ oe $T_A \cos PAB = T_B \cos PBA$ $\frac{2\lambda_A}{2} \times \frac{4}{5} = \frac{\lambda_B}{2} \times \frac{3}{5} \Rightarrow \lambda_B = \frac{8}{3} \lambda_A$ (ii) $T_A \sin PAB + T_B \sin PBA = mg$ $\frac{2\lambda_A}{2} \times \frac{3}{5} + \frac{8}{3} \times \frac{\lambda_A}{2} \times \frac{4}{5} = mg$ $\lambda_A = \frac{3}{5} mg$	B1 (AO 3.3) B1 (AO 1.1) M1 (Ao 3.1b) A1 (Ao 1.1) [4] M1 (Ao 3.1b) M1 (Ao 1.1) A1 (Ao 1.1) [3]	Hooke's law used for both strings Correct trig for both angles used Equating horizontal component s AG Both tensions resolved Vertical equilibrium; 3 forces Using tensions and $\lambda_B = \frac{8}{3} \lambda_A$		
		b	EPE = $\frac{\lambda_A \times 2^2}{2 \times 2} + \frac{\lambda_B \times 1^2}{2 \times 2}$ = mg	M1 (AO 3.4) A1 (AO 1.1) [2]	soi		



Question			Answer/Indicative content	Marks	Part marks and guidance		
		c	$T'_A = 0.3mg = \frac{\frac{3}{5}mgx_A}{2}$ $T'_B = mg + T'_A$ $1.3mg = \frac{\frac{8}{3} \times \frac{3}{5}mgx_B}{2}$ Distance is $1 + 2 + \frac{13}{8} + 2 = 6\frac{5}{8} \text{ m}$	M1 (AO 3.1b) M1 (AO 1.1) M1 (AO 1.1) A1 (AO 3.2a) [4]	Considerin g equilibrium of Q and using Hooke's law Considerin g equilibrium of P; must involve 3 terms Use of Hooke's law for upper string, with T'_A eliminated oe, e.g. awrt 6.63 m	($x_A = 1$) Or $T'_B = mg + 0.3mg$ soi ($x_B = \frac{13}{8}$)	
			Total	13			
17		a	$D - 140 - 850g \sin 7^\circ = 850 \times 0.4$ $P = D \times 12 \text{ oe soi}$ awrt 17 900 (or 17.9kW)	M1 (AO 3.3) M1 (AO 1.1) A1 (AO 1.1) [3]	NII, 4 terms, component of mg $D = 1495$ 17 942.06 W		
		b	$\frac{17\,942.06}{v} = 140 + 850g \sin 7^\circ$ $v = 15.5$ so the constant speed is 15.5ms^{-1}	M1 (AO 3.4) A1 (AO 1.1) [2]	FT their 17 942.06		
			Total	5			



Question			Answer/Indicative content	Marks	Part marks and guidance		
18		a	Components of initial vel: $6.5 \cos 30^\circ \downarrow$, $6.5 \sin 30^\circ \rightarrow$ Upwards component of final vel: $v_V = \frac{2}{3} \times 6.5 \cos 30^\circ$ $-2 = 2.5v_H - 2.5 \times 6.5 \sin 30^\circ$ $v_H = 2.45$ $v = \sqrt{v_V^2 + v_H^2}$, $\tan \theta = \frac{v_H}{v_V}$, oe Velocity has magnitude 4.48 ms^{-1} at 33.1° to the normal	M1(A03.1b) M1(A01.1) M1(A01.1) A1(A01.1) M1(A01.1) A1(A03.2a) A1(A03.2a) [7]	For correct use of either component (3.7527...) Use of impulse-momentum ; must involve horizontal components only For use of either trig or Pythagoras	Condone 33.2°	
		b	$\frac{\frac{1}{2} \times 2.5 \times (6.5^2 - 4.48^2)}{\frac{1}{2} \times 2.5 \times 6.5^2} \times 100$ Percentage loss is = 52.5	M1(A01.1a) A1(A01.1) [2]	Factor $\frac{1}{2} \times 6.5$ s of may be omitted Condone answers 52 and 53		
			Total	9			



Question			Answer/Indicative content	Marks	Part marks and guidance	
19		a	$= \frac{1}{2} \times 1.7 \times 0.5^2 + \frac{50 \times (3.2 - 1.2)^2}{2 \times 1.2}$ Initial energy 83.5 Minimum energy required to reach O is $1.7 \times 9.8 \times 3.2 + \frac{50 \times 1.2^2}{2 \times 1.2} = 83.3(12)$ So according to the model B reaches O	M1(A03.4) A1(A01.1) B1(A01.1) E1(A03.2a) [4]	May include a gravitational PE term	

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Question			Answer/Indicative content	Marks	Part marks and guidance		
		b	(i) Yes because work would have to be done against air resistance so more energy would be required to reach O and so the conclusion could change (ii) No because a string has no elastic energy after it goes slack so less initial energy would be required to enable B to reach O and so the conclusion does not change (iii) Either) The direction of projection has no effect on energy calculations. so (if nothing else interferes with the motion) the conclusion does not change Or If the downward motion means that the ball hits the floor some energy will be lost, so the conclusion could change Or any other sensible discussion about how the model might break down, e.g. the downward motion causing the spring to break or Hooke's law to fail, with an appropriate conclusion stated	E1(A02.4) [1] E1(A02.4) [1] E1(A02.4) [1]	Or work has to be done against the thrust as the spring goes into compression		

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Question			Answer/Indicative content	Marks	Part marks and guidance		
			Total	7			
20		a	$T \sin \theta = 3.5 \times 3^2 \times r$ $r = L \sin \theta$ $T \sin \theta = 3.5 \times 3^2 \times L \sin \theta$ oe $T = 31.5L$	M1(A03.3) B1(A01.1) M1(A01.1) oe A1(A01.1) [4]	Resolving tension and using NII Eliminating r AG working must be clear	may use $T \cos \phi$ instead or $r = L \cos \phi$	
		b	$T = \frac{75(L - 0.8)}{0.8}$	B1(A03.3) [1]	Correct use of Hooke's law	$(T = 93.75L - 75)$	
		c	$31.5L = 93.75(L - 0.8)$ $T = 38.0$ $L = 1.20$	M1(A01.1a) A1(A01.1) A1(A01.1) [3]	Solution of simultaneous equations		
		d	$T \cos \theta = mg$ 25.3°	M1(A03.4) A1(A01.1) [2]	Resolve vertically		
			Total	10			



Question			Answer/Indicative content	Marks	Part marks and guidance		
21		a	$WD = \int_0^4 (12x - 3x^2) dx$ Work done by the force = 32 J	M1(A01.1a) A1(A01.1) [2]	BC		
		b	Initial (=8.4 $KE = \frac{1}{2} \times 4.2 \times 2^2$) Work done = change in KE $\frac{1}{2} \times 4.2v^2 = 32 + 8.4$ Speed is 4.39ms ⁻¹	B1(A01.1) M1(A03.3) A1Ft(A01.1) A1(A02.2b) [4]	Need not be evaluated here Work-Energy principle used		
			Total	6			

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Question			Answer/Indicative content	Marks	Part marks and guidance	
22		i	Driving force $= \frac{23000}{v} \text{ or } \frac{23000}{8}$ $1200g \sin 4$ $\frac{23000}{v} - 800 - 1200g \sin 4 = 1200a$ $v = 8 \Rightarrow a = 1.05 \text{ m s}^{-2}$ $a = 0 \Rightarrow v_{\text{steady}} = 14.2 \text{ m s}^{-1}$	B1 B1 M1 A1 A1 [5]	N2L with either 3 or 4 terms 1.045553... 14.194585... Examiner's Comments The majority of candidates performed well on this question, showing a good understanding of the use of power in both situations. Where an error was seen, this was generally to omit the weight component in their solution.	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	WD by engine = 23000(10.1) WD against resistance = 800(109) Change in PE = 1200g(109sin 4) Change in $KE = \pm \frac{1}{2}(1200)(v^2 - 8^2)$ $23000(10.1) = \frac{1}{2}(1200)(v^2 - 8^2) +$ 800(109) + 1200g (109sin 4) $v = 12.5 \text{ m s}^{-1}$	B1 B1 B1 B1 M1 A1 A1 [7]	232300 87200 89416.638... Use of correct formula for KE to get a change in KE Use of conservation of energy, all terms present and no extra terms 12.522204... Examiner's Comments This part proved more difficult for a significant number of candidates. The main issues were to omit at least one energy term from a conservation of energy equation, to have incorrect signs with the terms also. As in previous years, some candidates duplicated the gravitational potential energy term by including the weight component down the slope in the resistance to motion.	
			Total	12		



Question			Answer/Indicative content	Marks	Part marks and guidance	
23			$T \cos 25$ $4000 = (T \cos 25)(75)$ $T = 58.8\text{N}$	B1 M1 A1 [3]	Seen in a WD equation $WD = F \times d$, must have component of T 58.8468... Examiner's Comments This question was well done by the majority of candidates, with only a minority not making an attempt to resolve the force T in the direction of motion.	
			Total	3		



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Question			Answer/Indicative content	Marks	Part marks and guidance		
24		i	$\frac{1}{2}m \times 0.7^2 = \frac{1}{2}mv^2 + \frac{24mg0.3^2}{2 \times 1.2} - mg \times 1.5$ Speed = 3.5 (ms ⁻¹)	M1 A1 A1 [3]	By energy; needs KE, PE and EE terms OR $\frac{1}{2}m \times 4.9^2 = \frac{1}{2}mv^2 + \frac{24mg0.3^2}{2 \times 1.2} - mg \times 0.3$ AG Adequate working, no errors	Allow wrong signs, missing '2'	

Examiner's Comments

Almost all candidates were able to do this part correctly. Some candidates went through an unnecessary extra step by first finding the speed of the particle when it had fallen 1.2 m from O and the string was not stretched. A small number of candidates ignored the initial speed of the particle, and a few others made sign errors in the energy equation and so were unable to prove the result legitimately.



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	One correct EE term involving x seen	B1		
			$\frac{1}{2}m \times 0.7^2 = \frac{24mg(x - 1.2)^2}{2 \times 1.2} + \frac{32mg(x - 1.5)^2}{2 \times 0.8} - mgx$	M1	Where x is distance below O	Energy equation with at least 1 KE, 1 PE and 1 EE term and values subst.
				A1	OR, where x is dist from T, $\frac{1}{2}m \times 0.7^2 = \frac{24mg(x+0.3)^2}{2 \times 1.2} + \frac{32mgx^2}{2 \times 0.8} - mg(x + 1.5)$	Alt left side: $\frac{1}{2}m3.5^2 + \frac{24mg0.3^2}{2 \times 1.2} - 1.5mg$
			$[48x^2 - 136x + 95 = 0]$	M1	Leads to $48x^2 + 8x - 1 = 0$	Dep M1 above
			1.25 (m) and 1.58 (m)	A1	Correct attempt to solve their 3 term quad.	
				[5]	$1\frac{1}{4} - 1\frac{7}{12}$	
					Examiner's Comments This question proved difficult, and candidates tackled it in a number of different ways. The approach which was most often successful was to equate the initial kinetic energy of the particle at O with the total energy in the spring and the string at a distance x m below O less the loss in potential energy. This approach leads directly to a quadratic equation which has the required distances as its roots. Other approaches were to define x as the extension of the string, or the reduction in length of the spring or as the length of the spring when compressed, and then find and solve a	



Question			Answer/Indicative content	Marks	Part marks and guidance	
					quadratic equation. Some candidates used the total energy of the system when the particle was at T as their starting point but they often missed one or other of the terms needed in forming the energy equation. There were some good attempts at this question where the candidate then went on to discount one of the solutions, and only gave one length as the answer. It is also possible to show that, after becoming attached to the spring, the particle performs simple harmonic motion, and then find the required distances. This approach was only rarely seen.	
			Total	8		



Question			Answer/Indicative content	Marks	Part marks and guidance		
25		a	$35g(5) - \frac{1}{2}(35)v^2 = \dots$ $\dots 35g(7 - 4\cos\theta)$ $v^2 = 4g(2\cos\theta - 1)$	M1(A03.3) A1(A01.1) A1(A01.1) A1(A01.1) E1(A02.1) [5]	Attempt at conservation of mechanical energy PE term on lhs KE term on rhs www; AG must show an intermediate step		
		b	When $\theta = 0^\circ$, $v_H = 2\sqrt{g}$ $v_v^2 = 2g(3)$ $v = 9.90$ so speed is 9.90 m s^{-1} $\theta = 50.8^\circ$ below the horizontal	B1(A03.3) B1(A03.3) M1(A03.4) A1(A01.1) A1(A01.1) [5]	Use of either $v^2 = v_H^2 + v_v^2$ or $\tan\theta = \frac{v_v}{v_H}$	6.260990... 9.8994949... 50.768479...	



Question			Answer/Indicative content	Marks	Part marks and guidance		
		c	Tension when $\theta = 60^\circ$ is 17.5g N $T - 35g \cos \theta = \frac{35}{4}(4g(2 \cos \theta - 1))$ $\theta = 33.6^\circ$	B1(A01.1) *M1(A03.1b) A1(A01.1) dep*M1(A01.1a) A1(A01.1) [5]	May appear anywhere in answer Attempt to use N2L radially with $a = \frac{v^2}{r}$ substituting 3×their(T) into their N2L attempt and solving for θ BC	N2L is Newton's second law 33.55730...	
			Total	15			



Question			Answer/Indicative content	Marks	Part marks and guidance		
26		a	$75g(6.2) + \frac{1}{2}(75)(2)^2 = 75g(1.2 - x) + \frac{10800x^2}{2(1.2)}$ $300x^2 - 49x - 255 = 0$	M1(A03.3) A1(A01.1) A1(A01.1) A1(A01.1) E1(A02.1) [1]	Attempt at conservation of energy GPE term(s) KE term EPE term www; AG must show intermediate step	GPE is Gravitational Potential Energy KE is Kinetic Energy EPE is Elastic Potential Energy	
		b	$x = -0.8438977... < 0$ so not valid $x = 1.01$	B1(A02.3) B1(A03.4) [2]	The negative value of x must be seen and not given as a final answer	1.007231...	
		c	e.g. $g = 9.8$ is not precise enough, use $g = 9.81$ e.g. the army recruit is not a particle, model as a rigid body e.g. the mat is not a single spring, model as multiple springs	E1(A03.5b) E1(A03.5c) [2]	For one assumption and a sensible refinement		
			Total	9			



Question			Answer/Indicative content	Marks	Part marks and guidance		
27		a	$\mathbf{a} = \frac{d\mathbf{v}}{dt} = 2t\mathbf{i} - \frac{10}{(2t+1)^2}\mathbf{j}$ $\mathbf{F} = m\mathbf{a} = 4t\mathbf{i} - \frac{20}{(2t+1)^2}\mathbf{j}$	M1(AO1.1) A1(AO1.1) A1FT(AO3.3) [3]	Differentiating one term of $\mathbf{v}$ correctly FT their $\mathbf{a}$		
		b	$\mathbf{F} \cdot \mathbf{v} = 4t(t^2 - 3) - \frac{100}{(2t+1)^3}$ Power is 208 W	M1(AO3.4) A1(AO1.1) A1(AO1.1) [3]	Attempt at scalar product 207.86282 ...	OR M1 at $t = 4$ $-(10\mathbf{i} - \frac{10}{9}\mathbf{j}) \cdot (13\mathbf{i} + \frac{5}{9}\mathbf{j})$	
		c	$T = \frac{1}{2}m\mathbf{v} \cdot \mathbf{v} = (t^2 - 3)^2 + \frac{25}{(2t+1)^2}$ Work done is 167 J	B1(AO3.4) M1(AO1.1) A1(AO1.1) [3]	$T(4) - T(2)$ 167.30864 2...	OR B1 for value of T at 2 or 4 M1 $T(4) - T(2)$	
			Total	9			



Question	Answer/Indicative content	Marks	Part marks and guidance	
28	Initial energy = $\frac{15 \times 0.5^2}{2 \times 0.6}$ $\frac{20x}{0.4} = \frac{15(0.5 - x)}{0.6}$ $x = \frac{1}{6}$ (m) $\frac{20 \times (\frac{1}{6})^2}{2 \times 0.4}$ or $\frac{15 \times (0.5 - \frac{1}{6})^2}{2 \times 0.6}$ seen KE = $3.125 - \frac{20 \times (\frac{1}{6})^2}{2 \times 0.4} - \frac{15 \times (0.5 - \frac{1}{6})^2}{2 \times 0.6}$ = 1.04167 (J) OR KE = $[3.125] - \frac{20x^2}{2 \times 0.4} - \frac{15(0.5 - x)^2}{2 \times 0.6}$ Differentiate OR complete the square Max KE when $x = \frac{1}{6}$ OR stationary when $3.125 = \frac{20x^2}{2 \times 0.4} + \frac{15(0.5 - x)^2}{2 \times 0.6}$ $x = 0$ or $\frac{1}{3}$ so max EPE when $x = \frac{1}{6}$ Or by SHM, $m\ddot{x} = \frac{15(0.5 - x)}{0.6} - \frac{20x}{0.4}$ $m\ddot{x} = -75\left(x - \frac{1}{6}\right)$ Amplitude, $a = \frac{1}{6}$ $\omega^2 = \frac{75}{m}$ Max KE = $\frac{1}{2}m\left(\frac{1}{6}\right)^2 \times \frac{75}{m}$	B1 M1A1 A1 B1 M1 A1 (M1) (M1) (A1) (M1A1) (A1) (M1A1) (A1) (B1) (B1) (M1)	3.125 (J) or $\frac{25}{8}$ x measured from B; other points possible OR $\frac{17}{30}$ from A, or $\frac{11}{60}$ from midpt Allow their $\frac{1}{6}$ Allow their $\frac{1}{6}$ = (3.125) - 0.69444 - 1.3888 Or $\frac{37.5}{36}$ or $\frac{25}{24}$ All terms present Allow slips B1 & B1M1A1 as above B1 & B1M1A1 as above Or other x	Or $1/3 + 0.6$ from C Not their $\frac{17}{30}$ or $\frac{11}{60}$ $\frac{25}{8} - \frac{25}{36} - \frac{25}{18}$ $\frac{75}{72}$ CTS $-37.5\left(\frac{1}{6} - x\right)^2 + \frac{37.5}{36}$ leads to max KE = $\frac{37.5}{36}$



Question			Answer/Indicative content	Marks	Part marks and guidance	
			$\frac{25}{24}$ J (or 1.04)	(A1)		Examiner's Comments The most common way of tackling this question was to find the equilibrium position for the system and the elastic energy in each string there. The maximum kinetic energy of B is the difference between and the elastic energy in the stretched string at the start, and the sum of the energy in each string at the equilibrium position. Many candidates did this successfully; though some candidates defining a variable other than the extension of one of the strings had difficulties. Other candidates used the fact that the motion of B would be simple harmonic, found the amplitude of the motion and then the greatest kinetic energy of B. Candidates using this method sometimes had a little difficulty dealing with the mass of B (which cancels out) and sometimes gave the greatest speed (in terms of mass) instead of what was required. A small number of candidates used the first method to find the amplitude of the motion, and then did the whole question by the SHM method, duplicating their earlier work; even so, these candidates were often completely successful. Another method was to find an expression for the kinetic energy of B in terms of an unknown length, and use differentiation or completing

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Question			Answer/Indicative content	Marks	Part marks and guidance	
						the square to find the answer.
			Total	7		



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Question			Answer/Indicative content	Marks	Part marks and guidance	
29		i	In equilibrium, $mg = \frac{24mge}{0.6}$	M1	Working essential, with $24mg$ and 0.6 used	Examiner's Comments This part was done correctly by the great majority of candidates.
		i	($e = 0.025$ m); OP = 0.625 m	A1	AG	
		ii	For Q falling 0.4 m $v^2 = 2g \times 0.4$	M1	Or $\frac{1}{2}mv^2 = mg \times 0.4$	Examiner's Comments Most candidates were able to do this correctly. A common error, however, was to assume that the potential energy lost by Q while falling was equal to the kinetic energy of the combined particle after they had coalesced. Many such candidates still claimed to have established the speed of 2.1 m s ⁻¹ .
		ii	$v = 2.8$ (m s ⁻¹) For combined particle	A1	Accept $\sqrt{(0.8g)}$	
		ii	$3m \times 2.8 = 4mV$	A1	Valid method shown – no slips	
		ii	$V = 2.1$ (m s ⁻¹)			
		iii	When a distance x below O $4mg - \frac{24mg(x - 0.6)}{0.6} = 4ma$	M1	Or $4mg - \frac{24mg(x + 0.1)}{0.6} = 4m\ddot{x}$	Allow sign error, m for $4m$ $4mg - \frac{24mg(x + 0.1)}{0.6} = 4m\ddot{x}$
		iii	$-10g(x - 0.7) = \ddot{x}$	A1	Accept a for $\ddot{x}$; $-10gx = \ddot{x}$	$-10g(x - 0.1) = \ddot{x}$
		iii	(SHM about) a point 0.7 m below O	B1		



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$2.1^2 = 10g(a^2 - 0.075^2)$	M1	Allow their ω ; any x	Use of $v = a\omega$ gains M1 Examiner's Comments Many candidates were able to find the centre of the motion, either from their attempt at showing simple harmonic motion or by considering the forces acting if the particle was at rest in that position. While most candidates were able to use Newton's second law to attempt to establish simple harmonic motion, only a minority did so correctly. The main error was to use a mass of m, rather than of $4m$, either once or twice in their equation. Partial credit was given to candidates attempting to use a correct form of equation to find the amplitude, but only those with a correct value of ω were able to get the correct value. A small number of candidates attempted to find the amplitude by considering energy, but most of these candidates failed to include potential energy in their calculations.
		iv	$0.4 = \frac{1}{2} 9.8t^2$; or $2.8 = 9.8t$	M1	Falling dist 0.4 m: 0.285714 s	Must be their 2.8
		iv	$0.075 = 0.225 \sin(\sqrt{98})$	M1	Travel 0.075m to centre: 0.0343287 s	Allow their values; allow cos
		iv	$\frac{1}{2}T = \frac{\pi}{\sqrt{98}}$	M1	Half period: 0.31734878 s	Allow for whole or part period
		iv	$0.1 = 0.225 \sin(\sqrt{98})$	M1	Travel 0.1m above centre: 0.046523 s	Allow their values; allow cos



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	$v = (0.225\sqrt{98} \cos(\sqrt{98} \times 0.046523)) (= 1.995)$ or $v^2 = 98(0.225^2 - 0.1^2)$	M1	Speed when string becomes slack 1.995307 m s ⁻¹	Allow their values; allow sin
		iv	$0 = 1.99 - 9.8t$	M1	Travel to highest point: 0.203602 s	Allow their values
		iv	Time = 0.888 s	A1	0.8875175 CAO	Dep M6
						Examiner's Comments Some candidates failed to attempt this part, and most who made an attempt did not consider all parts of the motion. This part was largely marked on the method used rather than accuracy, allowing 'follow through' from whatever values the candidates had for amplitude and ω. Most candidates who attempted the part were able to find the time it took Q to fall to the point where it coalesced with P. Candidates were next expected to find the time taken from this point to the centre of the simple harmonic motion. After this candidates had to realise that the combined particle performed simple harmonic motion only while the string was extended, so they needed to find the time to the point where the string next became slack, the speed of the particle at that point and then the time from there to the top of the motion. Fully correct solutions to this challenging question were rarely seen.
			Total	17		



Question			Answer/Indicative content	Marks	Part marks and guidance	
30		i	Use of $T = \frac{\lambda x}{l}$	M1	Used at least once	
		i	$T = \frac{10 \times 0.2}{0.4} + \frac{12 \times 0.1}{0.5}$	A1		
		i	$W = 7.4 \text{ N}$	A1	CAO AG	
		i	Use of $E = \frac{\lambda x^2}{2l}$	M1	Used at least once	
		i	$E = \frac{10(0.2)^2}{2 \times 0.4} + \frac{12(0.1)^2}{2 \times 0.5}$	A1		May see 0.5 + 0.12
		i	$E = 0.62(\text{J})$	A1	Examiner's Comments This part proved quite straightforward, though some candidates did not know the formula for the elastic potential energy stored in a stretched string, and a very small number did not work out the total elastic potential energy stored in the two strings.	
		ii	Use of $F = ma$ when further extension is x	M1*	Allow sign errors, 'm' wrong	OR, when total length of string is x
		ii	$7.4 - \frac{10 \times (x + 0.2)}{0.4} - \frac{12 \times (x + 0.1)}{0.5} = \frac{7.4}{g} a$	A1	'F' correct	$7.4 - \frac{10 \times (x - 0.4)}{0.4} - \frac{12 \times (x - 0.5)}{0.5} = \frac{7.4}{g} a$
		ii	$a = -\frac{49g}{7.4} x$	A1	Accept $a = -64.89...x$, $a = -\frac{2401}{37} x_{\text{oe}}$	$a = -\frac{49g}{7.4} (x - 0.6)$
		ii	SHM: $\omega^2 = \frac{49g}{7.4}$ (or $\frac{2401}{37}$ or 64.89189)	A1	Dep on all first 3 marks	SHM about $x = 0.6$, and ω^2 given
		ii	Use of $T = \frac{2\pi}{\omega}$	*M1	Must subst for their ω	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	period is 0.780 (secs) $\frac{2\pi\sqrt{37}}{49}$	A1	Allow if ω correct	0.77998
		ii	All subsequent motion is SHM because string does not become slack	B1	Justified at some point Examiner's Comments This part proved to be challenging. Most candidates sensibly considered a general position where each string was extended by a length x (m) from the equilibrium position. Other candidates used x to represent the total length of the stretched string, while a small number used other definitions. To gain full marks candidates were expected to state that $\ddot{x} = -\omega^2 x$ represented SHM. Where a candidate arrived at an equation of the form $\ddot{x} = -\omega^2(x - c)$ they were expected to explain what the constant represented; most did this appropriately. Candidates were also expected to show that P performs SHM throughout the subsequent motion; only a minority of candidates gained the mark for this. Few candidates had any difficulty with the period. A significant minority of candidates was unable to do this question; usually these candidates did not appreciate the need to use Newton's Second Law at a general point. It is possible to do this question by a consideration of energy at a	



Question			Answer/Indicative content	Marks	Part marks and guidance	
					general position, but only an extremely small number of successful solutions by this method were seen.	
			Total	13		



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Question			Answer/Indicative content	Marks	Part marks and guidance	
31		i	Conservation of energy	M1	Examiner's Comments	Need 4 terms; allow sign & trig errors
					Many candidates did this question completely correctly. a few made things more difficult than necessary for themselves by using a zero level for potential energy that was not the initial position, others made a sign error, or forgot about the kinetic energy of one of the particles. Quite a number of candidates made a sign error in the equation	
					$0.4g \sin \theta - R = \frac{0.4v^2}{a}$ and so did not find a correct expression for R ; a small number omitted the weight term. Almost all candidates knew that depended on putting $R = 0$, and those who had part (i) correct invariably gained both marks	
		i		M1		Both KE or both PE correct
		i	$\frac{1}{2}0.4v^2 + \frac{1}{2}0.6v^2 + 0.4ga \sin \theta - 0.6ga \theta = 0$	A1		completely correct
		i		M1	Attempt to find v^2 dep both earlier M1s	Allow with sign and trig errors
		i	$v^2 = 3.92a \quad (3 < 9 - 2\sin < 9)$	A1	AG	No errors
		i	$F = ma$ radially for P	M1*		Allow sign and trig errors
		i	$0.4g \sin \theta - R = \frac{0.4v^2}{a}$	A1		
		i		*M1	Manipulation attempted, leading to $a\theta + b \sin \theta$	Allow sign and trig errors
		i	$R = -4.7040 + 7.056 \sin \theta$	A1		2.352 $(-2 \theta + 3 \sin \theta)$



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	Using $R = 0$	M1	$0 = -4.704 (9 + 7.056 \sin \theta)$ Examiner's Comments Many candidates did this question completely correctly. A considerable number of candidates omitted this part.	
		ii	$(k =) \frac{2}{3}$	A1		Must be from correct expression in (i)
			Total	11		



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Question			Answer/Indicative content	Marks	Part marks and guidance	
32		i	$2.5g = 36.75 \text{ e}/3$	M1	P in equilibrium Examiner's Comments Most candidates correctly established the extension of the string and went on to find the speed of Q when it reached P and the common speed after that. A few candidates did not know how to use energy to find the speed of Q , and some tried to use energy to find the common speed.	Allow missing g
		i	$e = 2$	A1		
		i	$v^2 = 0^2 + 2g(3 + e)$	M1		
		i	$v = 7\sqrt{2}$	A1		May be implied by $v^2 = 98$
		i	$1 \times v = 3.5 \text{ V}$	M1		
		i	Combined speed = $2\sqrt{2} \text{ (ms}^{-1}\text{)}$	A1	AG	Convincing derivation, no errors
		ii	change in PE is $3.5gX$	B1	$34.3X$ Examiner's Comments This part was done extremely well, with only a few candidates going wrong by forgetting about the initial elastic energy or by making errors in working out the kinetic or potential energy terms. A small number of candidates used a wrong ' X ', though some of those managed to correct for this later and arrive at the correct equation.	
		ii	change in KE is $0.5 \times 3.5 (2\sqrt{2})^2$	B1	14	
		ii	change in EE is	M1		Allow sign errors / omission of 2;



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$36.75(X+2)^2/(2 \times 3) - 36.75 \times 2^2/(2 \times 3)$	A1		Allow 'x' or 'x + 5' for 'x + 2'; 2 terms or difference
		ii	Use conservation of energy	M1	$\frac{36.75(X+2)^2}{2 \times 3} = \frac{36.75 \times 2^2}{2 \times 3} + 3.5gX + \frac{3.5}{2}v^2$	Allow sign errors; at least PE, KE, EE term
		ii	$35X^2 - 56X - 80 = 0$	A1	AG	Convincing derivation, no errors may see $36.75X^2 - 58.8X - 84 = 0$
			Total	12		



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Question			Answer/Indicative content	Marks	Part marks and guidance	
33		i	By energy	M1*	Attempt at elastic energy	Allow M1 for $\frac{30y^2}{(2) \times 0.6} = kd$
		i	$\frac{30(d - 0.6)^2}{2 \times 0.6} = 48 \times d$	A1		$\frac{30x^2}{2 \times 0.6} = 48(x + 0.6)$
		i	$25d^2 - 78d + 9 = 0$	*M1	Get 3 term quadratic and attempt to solve	Allow 1 slip or $25x^2 - 48x - 28.8 = 0$
		i	or $30d^2 - 93.6d + 10.8 = 0$			
		i	$(d =) 3(\text{m})$	A1	Ignore $d = 0.12$, unless given as answer	$(x =) 2.4$ leading to $(d =) 3$
					Examiner's Comments	
					This question was done correctly by almost all candidates, though some made algebraic mistakes in moving from their original energy equation to a three term quadratic equation. A small number of candidates found the equilibrium position; while most then left this as their answer, a very small minority then went on to find the correct answer.	
		ii	Use $F = ma$	M1		Allow missing g , allow 1.3 or 0.6 to be omitted
		ii	$48 - \frac{30 \times (3 - 0.6 - 1.3)}{0.6} = (\pm) \frac{48}{g} a$	A1ft	ft their '3'	
		ii	$(a =) (+/-) 1.43$	A1	1.4291666	Using energy: $a = v \frac{dv}{dx} = \frac{g}{48} (50x - 72)$ M1A1



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	upwards	A1	Depends on a being right Examiner's Comments While this was generally done correctly, not all candidates remembered to take account of both the distance travelled of 1.3 m and the original length of 0.6 m in working out the extension of the string to be used. Just a few candidates either omitted the gravitational term completely or failed to make clear the direction of the acceleration.	
			Total	8		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
34			Use of $T = \frac{\lambda e}{l}$	M1	Attempt at one tension; allow use of x	allow 2/ for M1
					Examiner's Comments	
					Almost all candidates were able to do this question correctly. Many of them based their working on the length d given in the question; others used the length of one string or the other and added on either 0.4 m or 0.6 m at the end. A small number of candidates successfully worked with x_1 and x_2 as the lengths of the two strings. A few candidates used x for the length of each string, and failed to distinguish between them; other wrong answers included use of energy.	
				A1	$\frac{20(d-0.4)}{0.4}$ or $\frac{30(d-0.6)}{0.6}$	either term seen, accept in terms of x
			Weight = tension 1 + tension 2	M1		condone Wg and W/g
			(AW =) 1.5 (m)	A1	$100 = 50d - 20 + 50d - 30$	fractions and brackets removed
			Total	5		



Question			Answer/Indicative content	Marks	Part marks and guidance	
35		i	Use of energy equation at A and B	M1	3 terms needed $mg0.6\cos\frac{\pi}{6} = mg0.6\cos\theta + \frac{1}{2}mv^2$ Examiner's Comments This proved to be very straightforward for a large number of candidates who had been well trained and knew exactly what to do. Since the answer was given as a surd the final mark was not awarded to those who expressed $\sin(\pi/6)$ as a decimal in their working. However, quite a number of candidates made errors; some got the PE at A and B the other way round, and so had a negative quantity for v^2; others made a sign error in using $F = ma$; and some candidates had the particle start at T so that $\pi/6$ did not appear in the energy equation.	allow sign error, missing $m / g / r$
		i		A1		allow if θ replaced by $\varphi + \pi / 6$
		i	$F = ma$ radially	M1		allow sign error, missing m / g
		i		A1	$mg \cos \theta - R = \frac{mv^2}{0.6}$	
		i	Use of $R = 0$	M1	May be incorporated in previous step	
		i	$\cos TOB = \frac{\sqrt{3}}{3}$ AG	A1	Completely correct	not given if decimals used for angle.



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	Use of $\sqrt{3}/3$ in 'correct' equation in (i)	M1	$mg0.6\cos\frac{\pi}{6} = mg0.6 \times \frac{\sqrt{3}}{3} + \frac{1}{2}mv^2$ or $mg\frac{\sqrt{3}}{3} = \frac{mv^2}{0.6}$ Examiner's Comments Candidates who did part (i) correctly usually got this part right. Some who made errors in part (i), or made little attempt at part (i), also got this right by showing full correct working here.	equation must have gained M1 in (i) but allow restart here
		ii	1.84 (ms ⁻¹)	A1		
		iii	Use of $F = ma$ tangentially	M1	$mg\sin\theta = ma$ seen Examiner's Comments This proved a difficulty to some candidates, with some working out $g \cos \theta$, others not using an angle, and others not being able to work out $\sin \theta$ correctly.	allow missing m / g , – sign; allow M1 if angular accel found
		iii	8.00 (ms ⁻²)	A1		
			Total	10		

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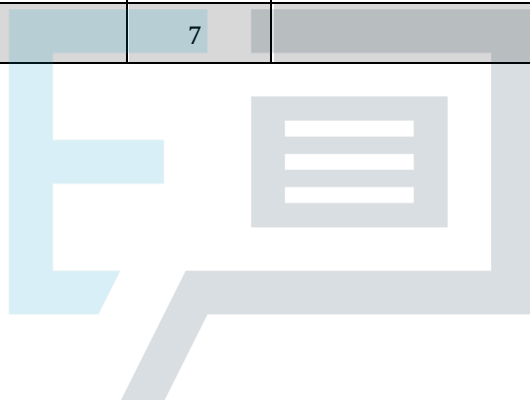
Question			Answer/Indicative content	Marks	Part marks and guidance	
36		i	Driving force = $\frac{23000}{v}$	B1		
		i	$\frac{23000}{v} - 600 = 1400(0.3)$	M1	Attempt at N2L with 3 terms; allow $D - 600 = 1400(0.3)$	
		i	$v = 22.5 \text{ ms}^{-1}$	A1	$v = 22.54901\dots$; allow Examiner's Comments Most candidates scored full marks for this part.	
		ii	$D - 600 - mg \sin 10 = 0$	M1	Attempt at N2L with three terms ($D = 2982.452998$); g needed	
		ii	$P = (cv(D))(12)$	M1	Use of $P = Dv$	
		ii	$P = 35.8\text{kW}$ or 35800 W	A1	$P = 35789.43597$ Examiner's Comments Again most candidates scored full marks, with only a minority making an error in the weight component.	
			Total	6		



Question			Answer/Indicative content	Marks	Part marks and guidance	
37		i	$Fr = 0.15(2.5g \cos 55)$	M1	Resolving perpendicular and use of $Fr = \mu R$	
		i		M1	Use of work done = friction $\times$ distance AB , using their friction; if combined with work done by other forces M0	
		i	Work done = 8.43N m	A1	Work done = 8.4315736... Examiner's Comments Most candidates made a good effort on this part with the only significant error seen was in how to deal with the distance of 4 m, in which some candidates read this as being the vertical height rather than a distance along the slope.	
		ii	$2.5g (4\sin 55)$	B1	PE term; 80.27690034	
		ii	$\pm \left(\frac{1}{2}(2.5)u^2 - \frac{1}{2}(2.5)(6.7)^2 \right)$	B1	Change in KE	
		ii	$\frac{1}{2}(2.5)u^2 - 2.5g(4\sin 55) - \frac{1}{2}(2.5)(6.7)^2 = cv(8.43)$	M1	Use of work – energy principle (4 terms); dimensionally correct	
		ii	$u = 10.8 \text{ m s}^{-1}$	A1	$u = 10.763678...$	
		ii	OR			
		ii	$+/-2.5a = -0.15(2.5g\cos 55) - 2.5g\sin 55$		Use of N2L with 3 terms ($a = -8.87074739...$: allow a positive)	
		ii		M1* A1		
		ii	$6.7^2 = u^2 + 2 \times 4 \times a$ $u = 10.8 \text{ m s}^{-1}$	M1 dep*	Method to find u using their deceleration, leading to $u > 6.7$ $u = 10.763678...$	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii		A1	Examiner's Comments By far the most common approach to this question by candidates was an energy method, which in most cases had all energy terms present but not always with the correct sign. The candidates who used a Newton's 2 nd Law approach usually answered the question correctly with an insignificant few failing to have all forces in their equation of motion.	
			Total	7		



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Question			Answer/Indicative content	Marks	Part marks and guidance	
38		i		M1*	Attempt at use of conservation of momentum	
		i	$4(8) + 3(-10) = 4v_A + 3v_B$	A1		
		i	$\frac{1}{2}(4)(8)^2 + \frac{1}{2}(3)(10)^2 - \frac{1}{2}(4)v_A^2 - \frac{1}{2}(3)v_B^2 = 121.5$	M1*		
		i		A1	Attempt at use of KE(before) – KE (after) = 121.5	
		i		M1 dep*	Obtaining quadratic equation in either v_A or v_B ($7v_B^2 - 4v_B - 416 = 0$, $28v_A^2 - 16v_A - 935 = 0$) and attempt to solve quadratic for either v_A or v_B	
		i	$v_A = -5.5$ ($v_A = 6.0714...$) so speed of A is $5.5 \text{ (m s}^{-1}\text{)}$	A1	cao; must be positive	
		i	$v_B = 8$ ($v_B = -7.428...$) so speed of B is $8 \text{ (m s}^{-1}\text{)}$	A1	cao; must be positive	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	Both particles are moving in the reverse direction to their original motion	A1	Or an equivalent statement consistent with their v_A and v_B; left and right not sufficient without a diagram; moving away from each other needs a diagram also Examiner's Comments A good source of marks for many candidates. A few made sign errors in the two equations but in most cases both the linear and quadratic equation were initially correct. The manipulation to obtain a quadratic in a single v often left much to be desired – errors when dealing with the fractions were abundant and several candidates thought that since $4v_A = 2 - 3v_B$, $4v_A^2$ could be replaced by $(2 - 3v_B)^2$. The necessity for answers to be positive was often not recognised, directions were sometimes indicated by arrows rather than a statement and statements, when given, more often than not included reference to left / right, or moving apart rather than the more explicit 'reverse direction to original'.	
		ii	$v_A - v_B = -e(8 - (-10))$	M1	Attempt at use of coefficient of restitution, right way round, v_A and v_B substituted	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$e = 0.75$	A1	Examiner's Comments The majority of candidates, who found a solution to part (i), were able to use the restitution equation, with only a few sign errors seen due to not taking into account the vector nature of the values used.	
			Total	10		



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Question			Answer/Indicative content	Marks	Part marks and guidance	
39		i	$D = 480/v$	B1	Use of $D = P/v$	
		i	$480/v - 60 = 0$	M1	Use of N2L with 2 terms to find v	
		i	$v = 8 \text{ m s}^{-1}$	A1	Examiner's Comments Most candidates scored full marks for this part.	
		ii	600×14.2	B1	WD by cyclist (8520 J)	
		ii	$\frac{1}{2}(80)(9.4^2 - 8^2)$	B1ft	ft v from (i); KE gained (974.4 J); may be implied in energy equation	
		ii	$8520 = 974.4 + 60d$	M1	Attempt at energy equation with all terms	
		ii	$d = 126 \text{ m}$	A1	Exact 125.76 Examiner's Comments This was not approached well by the majority of candidates. This is in fact an energy type problem, with many candidates assuming constant acceleration instead – a few of these gained B for calculating the work done, usually amongst some completely wrong working. A small number gained only the B1 for the change in KE. Even those who used an energy approach sometimes omitted one of the energy terms.	
			Total	7		



Question			Answer/Indicative content	Marks	Part marks and guidance	
40			$P/16$ or $P/25$ seen	B1	Use of Driving force = P/v	
			$P/16 - R = 1500(0.3875)$ or $P/25 - R = 1500(0.2)$	M1	Attempt to use N2L once, can be with D at this stage	
			$P/16 - R = 1500(0.3875)$ and $P/25 - R = 1500(0.2)$	A1	Both equations correct, may have consistent wrong sign with R	
				M1	Attempt to solve for P or R	
			$P = 12\,500$	A1	Allow 12.5 kW	
			$R = 200$	A1	Allow if -ve, and justified why +ve (eg. comment on direction)	
					Examiner's Comments	
					The majority of candidates completed the usual approach of using Newton's second law to produce two simultaneous equations successfully. However a minority didn't progress much further by failing to replace their driving force with $P/16$ and $P/25$. The solving of the simultaneous equations presented the usual opportunities for algebraic slips from the weaker candidates, with sign errors cropping up.	
			Total	6		



Question			Answer/Indicative content	Marks	Part marks and guidance	
41		i	$v^2 = 5^2 + 2g(1.6)$	B1	Complete method to find v (= 7.507...)	
		i	$0.7 \times 7.507... (= 5.255...)$	B1	$0.7 \times cv(v)$, but not $cv(v) = 5$; may be seen in (ii)	
		i	$(0.7 \times cv(v))^2 = 2gh$	M1	Complete method to find h , with final speed 0; allow $cv(v) = 5$ for method	
		i	$h = 1.41 \text{ m}$	A1	Exact 1.409 Examiner's Comments The vast majority of candidates scored full marks here. Of those who didn't, the common error was to use 5 m s^{-1} as the speed when hitting the ground.	
		ii		M1	Change in momentum found, with relevant velocities ie $cv(v)$ and $0.7 \times$ $cv(v)$ but not $cv(v) = 5$	
		ii	$0.2(7.507...)(0.7) - (-0.2)(7.507...)$	A1ft	This may be negative; ft on their v found in (i)	
		ii	Impulse = 2.55 N s, upwards	A1	(2.5524...) Must have direction also. Examiner's Comments Many candidates who were able to answer (i) correctly went wrong here. The common error was not to deal correctly with the vector aspect of the momentum of the sphere with a difference of two scalar quantities being seen significantly often.	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii		M1	Change in energy found, all energy terms needed and no extra terms This may be negative	
		iii	$0.2(9.8)(1.6) + \frac{1}{2}(0.2)(5^2) - 0.2(9.8)(cv(h))$ OR $\frac{1}{2}(0.2)(7.507\dots^2) - \frac{1}{2}(0.2)(0.7 \times 7.507\dots)^2$	A1ft		
		iii	Loss of energy = 2.87 J	A1	(2.87436 exact); art 2.87; allow -2.87 Examiner's Comments This was done relatively poorly. The problem was either through not considering all the energy terms needed (ignoring PE or KE) or using far too many (bringing in the 7.507 and/or the 5.255).	
			Total	10		



Question			Answer/Indicative content	Marks	Part marks and guidance	
42		i	Driving Force = $10000/20$ (= 500)	B1		
		i		M1	Attempt at N2L with 3 terms	
		i	$cv(10000/20) - 1300 + 800g\sin\alpha = 0$	A1		
		i	$\sin\alpha = 5/49$	A1	AG at least one more line of correct working (at least e.g. $-800+800g\sin\alpha=0$); allow verification (e.g. $500 - 1300 + 800 = 0$)	
					Examiner's Comments	
					As this was a given answer, candidates needed to have a convincing solution to arrive at the final answer and, on the whole, they did.	
		ii	$800(22.1)g\sin\alpha$	B1	Work done against weight; Need a value for $\sin\alpha$ or α	
		ii		M1	Total work done, 3 terms needed	
		ii	$800(22.1)g\sin\alpha + 1300(22.1) + \frac{1}{2}(800)(8^2)$	A1	Need a value for $\sin\alpha$ or α ; (72010 J)	
		ii		M1	Time = work done(from at least one correct energy term) / power	

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$t = 3.6(0) \text{ s}$	A1	'Exact' is 3.6005 Examiner's Comments This proved to be the hardest question on the paper. The solution to this question was to consider energy and many efficient solutions were seen. However there were many examples seen of candidates wrongly assuming that either speed or acceleration was constant. Candidates should be reminded that the only constant in this type of situation is power.	
			Total	9		

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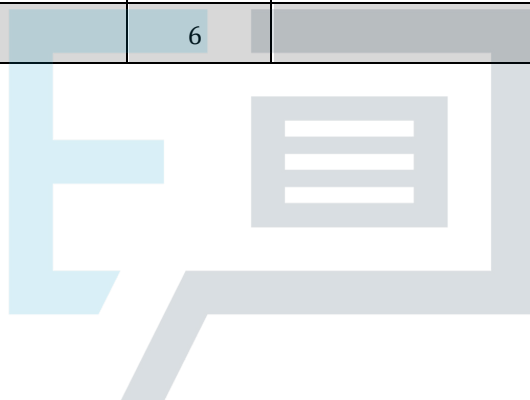
Question			Answer/Indicative content	Marks	Part marks and guidance	
43		i	$0.75 \times g \times 8$	M1	Weight $\times$ distance	
		i	58.8 J	A1	Allow -58.8 Examiner's Comments The majority of candidates answered this question well. Only a minority attempted to use kinetic energy to find the loss in potential energy.	
		ii		*M1	Attempt at change in KE	
		ii	$+/-(\frac{1}{2} \times 0.75 \times v^2 - \frac{1}{2} \times 0.75 \times 2^2)$	A1		
		ii	$\frac{1}{2} \times 0.75 \times v^2 - \frac{1}{2} \times 0.75 \times 2^2 = 58.8$	dep*M1	Equate their change in KE to their PE from (i)	
		ii	$v = 12.7 \text{ m s}^{-1}$	A1		
		ii	OR $a = g \sin \theta$	B1	θ is angle of slope to horizontal.	
		ii	$s = \frac{8}{\sin \theta}$	B1		
		ii	$v^2 = 2^2 + 2 \times g \sin \theta \times \frac{8}{\sin \theta}$	M1		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$v = 12.7 \text{ m s}^{-1}$	A1	Not $a = g$, not $s = 8$ Examiner's Comments This part was not answered well by a significant number of candidates. It was common to see the initial velocity of the particle being ignored in the energy equation and candidates finding a new wrong speed of 12.5 m s^{-1} . Some candidates then thought they could deal with the initial speed by adding it to their wrong value.	
			Total	6		



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Question			Answer/Indicative content	Marks	Part marks and guidance	
44		i	20000/32	B1		
		i	$R = 20000/32$	M1		
		i	$R = 625 \text{ N}$	A1	cao Examiner's Comments Generally was answered well by candidates. Examiners were pleased when candidates gave a clear argument that, at constant speed, resistance was equal to driving force.	
		ii		M1	Using Newton 2, all forces used.	
		ii	$F + 1500g\sin 2 - 625 = 1500 \times 0.1$	A1ft	ft their R from (i) SC $F - 1500g\sin 2 - 625 = 1500 \times 0.1$	
		ii	Power = $32 \times F$	M1	Using their F .	
		ii	Power = 8380 W or 8.38 kW	A1	8383.27.... SC 41200 W or 41.2 kW (41216.7...) Examiner's Comments In which the driving force had to be calculated using Newton's 2nd Law, was done very well although it was common to see the weight component (or even the acceleration term) missing from the resolution parallel to the plane. The majority were able to multiply their driving force by the speed to find a power.	
			Total	7		



Question			Answer/Indicative content	Marks	Part marks and guidance	
45		i		M1	Force component x distance	
		i	$18\cos 15^\circ \times 6$	A1		
		i	104 J	A1	Examiner's Comments This question was well answered. Only a small minority of candidates failed to resolve the force.	
		ii	$18\cos 15^\circ \times 6/5$ or $\text{ans(i)}/5$	M1	Force component x distance/5	
		ii	20.9 W	A1	Allow 20.8 Examiner's Comments Again well answered. Candidates either used their answer to (i) divided by time or used force component multiplied by speed with equal success.	
			Total	5		



Question			Answer/Indicative content	Marks	Part marks and guidance	
46		i	$DF = 15000/15$	B1		
		i		M1	N2L, 3 terms and attempt at DF.	
		i	$DF - k \times 15^{1/2} = 1500 \times 0.4$	A1	Numerical DF	
		i	$k = 103$	A1	Allow $80\sqrt{15}/3$	
					Examiner's Comments	
					The majority of candidates knew how to convert the power to a driving force and then use it in a Newton's second law equation successfully.	
		ii		M1		
		ii	$P/30 = k30^{1/2}$	A1	Using $cv(k)$	
		ii	$P = 17000W$	A1	Allow $17(.0)kW$, $16900W$, $16.9kW$, $12000\sqrt{2}W$	
					Examiner's Comments	
					This question was well done by the majority. Only a small minority attempted to use the acceleration given for part (i).	
			Total	7		

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